

6. Two common types of ripple counter are ripple up counter and ripple down counter. The circuit shown in Figure 1-1 is a 2-bit ripple (up) counter.

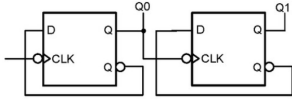


Figure 1-1

注意，下一级触发器的时钟有个圈是下降沿触发，该行波计数器为异步触发

假设从00开始计数

Q ₁	Q ₀
0	0
0	1
1	0
1	1

↓ negedge

从计数行为来看是加法计数器。

3. To design a "001010" serial sequence generator by shift registers, (4) flip-flops are required at least.

[滑动法]

1. 设定一个 n 值 (比如先试 $n = 3$)。
2. 把原序列写两遍连起来，保证循环能接上: 001010001010...
3. 用长度为 n 的框去框住前 n 位，记录下来。
4. 框往右移动1位，记录下新的 n 位。
5. 直到把整个循环移完，看记录下来的所有“框”里，有没有出现完全相同的二进制数。
6. 如果有重复: 说明 n 个触发器不够，因为电路到了这个重复的状态时，它“失忆”了，分不清下一步该输出0还是1。
7. 如果全都不重复 (唯一): 说明 n 个触发器就足够了!

比如 001010:001010

若 $n=3$, 那么两次框住不会会出现

001, 010, 101, 010, 100, 000, 010.

出现相同二进制数 那么对010这个位模式而言，
下位是1或者0，由于 shift register 的外部
电路为组合逻辑，不能存储状态，所以 $n=3$ 不行
(状态有重复)

在 $n=4$ 时，循环序列变为:

0010 - 0101 - 1010 - 0100 - 1000 - 0001 - 0010 - ...

不存在重复状态，所以可以。

3.

7. A 74×194 can be used as a sequence detector. When Z shown in Figure 1-2 is asserted, the sequence detected is (10110).

74x194, 双向Shift Register

$S_1, S_0 = 01$ 的时候右移, 即:

$$[Q_A, Q_B, Q_C, Q_D] \leftarrow [R_{in}, Q_A, Q_B, Q_C]$$

Z=1 满足 $Z = Q_D \cdot Q_C \cdot \bar{Q}_B \cdot Q_A \cdot \bar{R}_{in}$
所以此时, 寄存器存储的是

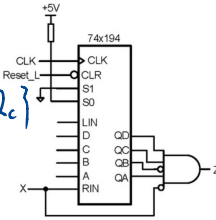


Figure 1-2

$$Q_A Q_B Q_C Q_D = 1011, \text{ 且 } R_{in} \text{ 为 } 0$$

根据串行输入的方式, 可以推理出最先检测到的是 Q_D 位置的值 (Shift Register 右移时的 LSB)

$\therefore$ Sequence be detected is: 11010

4.

4. A 4×4 unsigned binary multiplication can be built with a () ROM.

- A. 128×4 B. 256×4
C. 128×16 D. 256×8

4x4 的无符号乘法器, 输入为 8 位 ($A_3 A_2 A_1 A_0 \times B_3 B_2 B_1 B_0$)

所以 ROM 的地址是 8 位,

4x4 乘法输出为 $4+4=8$ 位, 所以 ROM 存储 8 位数据

容量为 $2^8 \times 8 \rightarrow$ 每个数 8 位

↑
8 位地址, 要存储 2^8 个数

5.

1. The state/output table of an FSM circuit is shown in Table 1. X is input and Y is output. According to the table, the correct statement is (C).

Table 1 State/output table			
S	X		Y
	0	1	
A	A	C	0
B	B	A	0
C	C	B	1
D	A	A	0
S*			

- A. Mealy machine, 3 valid states B. Mealy machine, 4 valid states
C. Moore machine, 3 valid states D. Moore machine, 4 valid states

在该状态机运行时，D是不可达的，所以实际只有3个有效状态

观察题中的状态转移表（Table1），我们重点看 次态 S*（Next State）的两列（即 X = 0 和 X = 1 时对应的状态）：

- 当当前状态为 A 时，次态为：A 或 C
- 当当前状态为 B 时，次态为：B 或 A
- 当当前状态为 C 时，次态为：C 或 B
- 当当前状态为 D 时，次态为：A 或 A

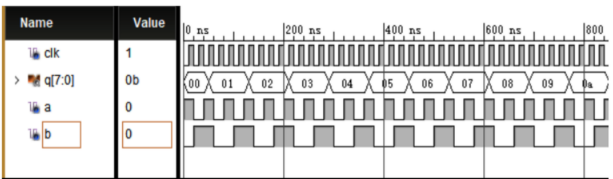
关键分析

1. 状态 D 是不可达的（Unreachable）：没有任何一个状态（包括 D 自己）在输入 X = 0 或 X = 1 时能够转移到状态 D。也就是说，D 永远不会作为“下一个状态”出现。
2. 正常工作状态循环：无论电路的初始状态是什么，只要电路开始运行，状态就会进入 A、B、C 这三个状态构成的循环圈中，再也无法回到或进入状态 D。

因此，状态 D 属于无效（或不可达）状态。该有限状态机（FSM）在实际稳定运行中只有 A、B、C 这 3 个有效状态。

6. 代码改错：

得分 III. We need to design a verilog hdl program to realize a special counter. The correct simulation result is shown below. But there are several errors in the main module shown below. Please find the errors and correct them. (10')



```

`timescale 1ns / 1ps //1
module p1(clk,a,b,q) //2
input clk; //3
input a; //4
input b; //5
output q[7:0]; //6
reg [7:0] q=0; //7
always @( clk ) //8
begin //9
    if (a & b) //10
        q++; //11
    else //12
        q=0; //13
end //14
endmodule //15

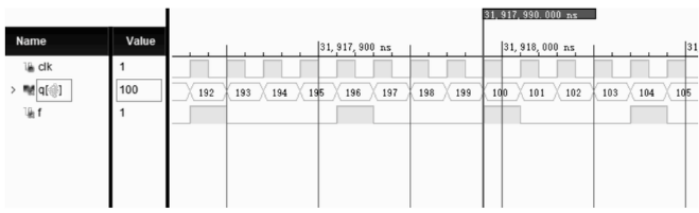
```

Line (2) is error. Modified to: (module p1(clk,a,b,q);)
Line (6) is error. Modified to: (output [7:0] q;)
Line (7) is error. Modified to: (always @(posedge clk))
Line (11) is error. Modified to: (q <= q+1;)
Line (13) is error. Modified to: (q <= q; 或 q = q)

7. Verilog
 2. 修改



III. A verilog HDL module realizes a special binary counter. The count value is from 100 to 199. The output f will be asserted when the count value is an integer multiple of 4. The correct simulation result is shown below.



We wrote this module code as shown below. But there are several errors. Please find the errors and correct them. (10')

```

module p1(clk, q, f); //1
input clk; //2
output [3:0] q; //3
output f; //4
reg [7:0] q = 0; //5
assign f = q[1] & q[0]; //6
always @ (posedge clk) //7
begin //8
    if (q = 199) //9
        q <= 0; //10
    else //11
        q <= q + 1; //12
end endmodule //13
    
```

Please write the number of error line and correct it. You can append lines below, if you think not enough.

- Line (3) is error. Modified to: (output [7:0] q;)
- Line (5) is error. Modified to: (reg [7:0] q = 100; (100~199 都对))
- Line (6) is error. Modified to: (assign f = ~q[1] & ~q[0]; 或 assign f = ~(q[1] | q[0]);)
- Line (9) is error. Modified to: (if (q == 199))
- Line (10) is error. Modified to: (q <= 100)
- Line () is error. Modified to: ()
- Line () is error. Modified to: ()

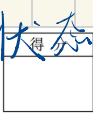
8

7. If the clock input is connected to a 10MHz digital signal, which of the following circuits can output a 1MHz signal with 50% duty cycle? (B)

- A) a self-correcting 10-bit ring counter with a single circulating 1
- B) a self-correcting 5-bit Johnson counter
- C) a free-running decade counter (十进制计数器)
- D) a 4-bit LFSR counter

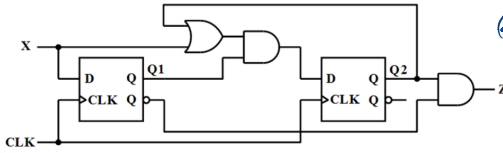
模10
 有效状态数为2n=10,任意一个输出端在10个周期内5个周期高电平,5个周期低电平

9. 状态机分析 (Moore 型, 输出和状态有关)



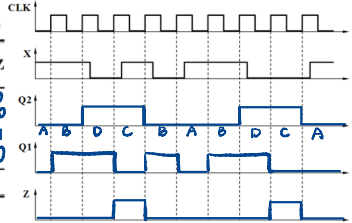
III. Analyze the sequential-circuit as shown below. [18']

1. Write out the excitation equations and output equation. [6']
2. List out transition/output table and draw a state diagram. [5']
3. Assume the initial state $Q_2Q_1=00$, draw the timing diagram for Q_2, Q_1 and Z . [5']
4. Indicate the logic function of the circuit. [2']



Transition/output table

S	X		Z
	0	1	
A 00	00	01	0
B 01	00	01	1
C 10	00	01	1
D 11	10	11	0
Q2Q1	Q2*Q1*		



① 激励方程 (写出口端的逻辑表达式)

$$D_2 = Q_2^* = (X + Q_2)Q_1$$

$$D_1 = Q_1^* = X$$

② 输出: $Z = Q_2 \bar{Q}_1$

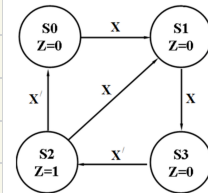
Assume that $A=2'b00$

$B=2'b01$

$C=2'b10$

$D=2'b11$

状态图 (State diagram)

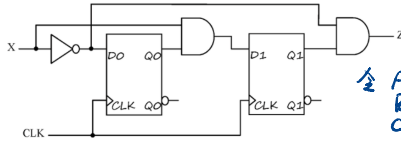


行为: Moore 型, 进行 110 的序列检测

10. 状态机分析 (Mealy 型, 输出和输入以及当前状态有关)

III. Analyze the clocked synchronous state machine as shown below. (15')

- (1) Write out the excitation equations, transition equations and output equation. [2']
- (2) List out the transition/output table. [4']
- (3) Assume the initial state Q_1Q_0 is 00, draw the timing diagram for Q_0, Q_1 and Z . [7']
- (4) Indicate the function for the logic circuit. [2']



全 $A=00$
 $B=01$
 $C=10$
 $D=11$

(2) 每格 0.5 分, 共 4 分。

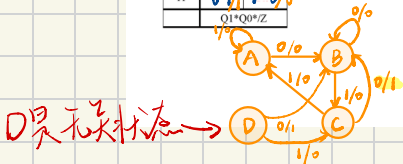
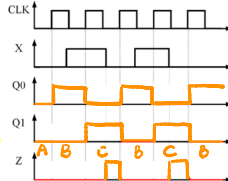
Transition/output table

Q1Q0	X		Z
	0	1	
00	0/0	0/0	
01	0/1	0/0	
10	0/1	0/0	
11	0/1	0/0	
	Q1*Q0*/Z		

$$D_1 = Q_1^* = \bar{X}$$

$$D_0 = Q_0^* = X \cdot Q_0$$

$$Z = \bar{X} \cdot Q_1$$



无关状态 →

function: 可重叠的 010 序列检测

11. 根据描述设计状态机

VI. Design a 2-bit Gray-code up/down counter with two D Flip-flops and some gates. Q1Q0 represent the counter value in Gray-code. M is the up/down select input. When M=1, the counter counts in ascending Gray-code order and then the output Z will be 1 when counter value reaches 3. When M=0, the counter counts in descending Gray-code order and then the output Z will be 1 when counter value reaches 0. (15')

- (1) List out the transition/output table. [4']
- (2) Fill out the Karnaugh maps. [3']
- (3) Write out the transition equations, excitation equations and output equation. [4']
- (4) Draw the logic circuit diagram. [4']

【参考答案】

Transition/output table		
Q1Q0	M	
	0	1
00	10/1	01/0
01	00/0	11/0
10	11/0	00/1
11	01/0	10/0
	Q1*Q0*/Z	

格雷码计数器:

M=1: 00-01-11-10 (ascending)

M=0: 00-10-11-01 (descending)

卡诺图:

Q_1^*

M \ Q0	00	01	11	10
0	1	0	0	1
1	0	1	1	0

$$Q_1^* = \bar{M} \cdot \bar{Q}_0 + M \cdot Q_0$$

Q_0^*

M \ Q0	00	01	11	10
0	0	0	1	1
1	1	1	0	0

$$Q_0^* = \bar{M} \cdot Q_1 + M \cdot \bar{Q}_1$$

Z

M \ Q0	00	01	11	10
0	1	0	0	0
1	0	0	0	1

$$Z = \bar{M} \bar{Q}_1 \bar{Q}_0 + M Q_1 \bar{Q}_0$$